Do central bank reserves stabilize markets? A fixed effects analysis between CDS and Forex Reserves

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Abstract

In this paper, we examine the relationship between sovereign credit default swaps (CDS) and foreign currency reserves. This study observes 9 emerging market economies and 11 advanced market economies covering high frequency (i.e., daily) data spanning from June 1, 2015 to June 30, 2023. We started with a linear regression and found that a 1 billion USD increase in reserves leads to a decrease in Sovereign CDS prices by 0.0323 USD. After accounting for control variables such as GDP and trade surplus, changing to a fixed effects regression, and transforming sovereign CDS prices and foreign reserve nominal values into logarithmic data, we find that reserves and CDS price have a strong inverse relationship. Increasing foreign reserves by 1% decreases sovereign CDS prices by 0.558%. Our analysis builds on the existing literature by including both Advanced and Emerging market economies, as well as utilizing up-to-date data.

1. Introduction

There has long been a debate regarding the optimal level of central bank intervention in foreign exchange markets. Friedman (1953) famously argued in "The Case for Flexible Exchange Rates" that currency exchanges should be free floating and highly stable. This school of thought, which argues against interventionist policy, does not see a need to maintain high levels of foreign currency reserves. It seems this belief did not prevail in practice. Foreign currency reserves as a percentage of GDP have soared from 5% in the 1980s to about 25% in 2010 (Uz Akdogan 2019), prompting academics and central bankers alike to look more closely at this metric.

In trying to understand the rationale of this recent expansion in reserves we look at credit markets. Perhaps changes in reserves is an effective way to alter perceived risk, leading to cheaper borrowing costs. Higher risk premiums can drown a country in high interest rates expenses, whilst a low-risk premium can allow governments to borrow cheaply, propelling investment. We identified sovereign credit default swaps (CDS), the most popular being the 5-year CDS contract, as the best indicator to measure a country's credit risk. This financial derivative acts as a form of insurance against the risk of a country defaulting on its debt obligations. The buyer of a CDS pays a periodic fee, called the spread, to the seller. The spread reflects the perceived risk of default, with higher spreads implying a higher credit risk.

This paper begins with a brief literature review of current knowledge about the key variables of our study and relationship between them. First, we consider the legitimacy of sovereign CDS as a metric (section 1.1); then, the relevance of foreign currency reserves (section 1.2); and finally, the relationship between both variables (section 1.3).

1.1. Why we chose CDS and its importance

The literature surrounding sovereign CDS is mostly based on its use as an indicator of a country's risk. For example, CDS is used to address the *forward premium puzzle*, where currencies with high interest rates tend to appreciate more than predicted by the uncovered interest rate parity (UIP). Coudert and Mignon (2013) sample the carry trade between 18 emerging market currencies between 2005 and 2010 and find that utilizing sovereign CDS as a proxy for sovereign risk largely explains this discrepancy. This is in line with Foroni et al. (2018), and Wang & Liang (2024).

Beyond the forward premium puzzle, Rodríguez et al. (2019) finds that sovereign CDS have strong predictive power for sovereign credit events, significantly outperforming sovereign credit ratings. Lastly, Longstaff et al. (2011) offered a unique, but not necessarily opposing viewpoint to the previously mentioned works. He argued that up to 64% of the variation in sovereign credit spreads are primarily explained by global macroeconomic factors and liquidity conditions. In sum, the research to date indicates that CDS is a good indicator of sovereign risk.

1.2. Why we chose foreign reserves as a measure of intervention

For some time now, measuring central bank intervention in markets has been a point of contention for academics. Despite its importance in understanding exchange rates, central banks have been unwilling to provide researchers with official data concerning such interventions. This difficulty in accessing central bank intervention data highlights the importance of Uz Akdogan (2019). The author argues that "data from official foreign reserves provide enough evidence for detecting policy changes and reveal central bank intervention."

Although the methods utilized by Akdogan (2019) to isolate for central bank intervention are beyond the scope of this paper, his research provides strong evidence for using foreign reserves as the primary variable in this study. It also supports our decision to use GDP and trade surplus as control variables in the study and use a fixed effects regression method of analysis.

1.3. The existing relationship between foreign reserves and CDS

There is debate in the literature about the relationship between foreign currency reserves and sovereign CDS. Can et al. (2023) collects Turkish data from 2010 to 2022 and finds that there is a weak unidirectional negative correlation from CDS to central bank reserves (CBR). Suggesting that higher borrowing costs, reflected in higher CDS premiums, may drive the use of reserves to meet financial obligations. Kartal et al. (2023) also collects Turkish data from 2004 to 2021. They find that there exists a bidirectional link between CBR and CDS spreads. Akçelik & Fendoğlu (2019) analyze seven emerging market economies between 2005 and 2017 to find that a 1% decrease in reserves increase spreads by 1.77% when global risk appetite is not accommodative and decrease spreads by 1.40% when global risk appetite is accommodative.

On the other hand, Ho (2016) studies the long and short-run determinants of sovereign CDS spreads for eight countries from Q4 2008 to Q2 2013. They find that international reserves

have the largest impact on reducing CDS spreads. Demonstrating that a 1% increase in reserves leads to a decrease of the CDS spread by about 2.73%.

Observing these mixed results, we hypothesized that Forex reserves would have a negative correlation with sovereign CDS spreads. We expanded upon the literature by including up-to-date data on both advanced and emerging market economies. We performed a fixed effects regression to find that foreign currency reserves have an inverse relationship with sovereign CDS prices. When taking the logarithmic form of both reserves and swaps, as well as accounting for controls, a 1% increase in reserves leads to a -0.558% decrease in spreads. This result is both statistically and economically significant.

2. Data

This study utilizes data on multiple economic indicators to isolate the effects of foreign currency reserves on sovereign CDS prices. We collected data on GDP, trade surplus, CDS prices, and foreign currency reserves. Each variable was collected separately and then combined into a single panel dataset. Additionally, each variable was collected for the same 20 countries from June 1, 2015 to June 30, 2023. These include 11 developed countries: Canada, Japan, South Korea, United Kingdom, USA, Italy, Spain, France, Israel, Australia, and Germany and 9 developing nations: Mexico, Brazil, China, South Africa, Indonesia, Egypt, India, Turkey, and Saudi Arabia. Each data entry represents a single day for a single country, represented in Figure 1. Any lower frequency data collected, e.g. yearly, was duplicated to match the daily frequency. For example, each observation for Spain in 2016 has the same yearly GDP value.

The Sovereign CDS data was gathered through *investing.com*, a financial services platform. This dataset includes daily prices in USD for the working weekdays in the mentioned period. The mean price was 118.58 with a large standard deviation of 152.810, suggesting high variability between countries. The Forex reserves was collected from *Trading Economics*, a macroeconomic data hub. This was weekly and in millions USD. The mean reserves were 4,923 billion USD with a large standard deviation of 7,867 billion USD, as seen on *Table 1.* This supports the hypothesis of the existence of high variability between the countries in the dataset.

GDP for each country was collected from FRED, an online database managed by the Federal Reserve Bank of St. Louis. The data was yearly and in USD. Trade surplus was collected

for each country from the World Bank. This was yearly data in percentage of GDP. Each of the collected datasets was merged into a single Panel Data Excel file before being used in the analysis of this paper. The result was a dataset with 40,129 observations.

3. Regression Analysis

3.1. Simple Linear Regression

Specification 1 estimates the relationship between foreign currency reserves and Sovereign CDS prices through a simple linear regression modeled by the equation:

$$CDS_{i,t} = \beta_0 + \beta_1 Reserves_{i,t} + e_{i,t}$$

We use robust standard errors to account for any heteroskedasticity. The result showed that an increase in reserves by 1 billion USD leads to a decrease in Sovereign CDS prices by 0.0323 USD. This result was statistically significant with a 95% confidence interval of (–0.0342, -0.0305). Additionally with a t-statistic of 54.89 we reject the null (H_0: $\beta_1 = 0$) that there is no link between foreign currency reserves and CDS prices. However, this result is economically insignificant as CDS prices ranged in our data from 5.46 to 1836.25 USD. Additionally, some countries had as little as a billion in reserves. Therefore, doubling these reserves for a decrease of only 0.03$ would not be an effective way to affect CDS prices.

While this result suggests a limited negative relationship between the variables, there are threats to the validity of this regression. It fails to control for any variables that may be influencing CDS Prices. By introducing controls, we isolate the effect of the variable of interest on the dependent variable. This will reduce bias in the estimator by removing confounding variables and preventing omitted variable bias. Another issue with the simple linear regression model is the lack of control for country-specific characteristics that could be influencing CDS Prices. These are time-invariant characteristics which can be known or unknown. For example, the effects of having a pegged currency which would be constant over the time period. It isolates the effect of time-varying independent variables on the dependent variable by removing the influence of entity-specific characteristics. Finally, the dataset has large variability between countries as seen in CDS prices and reserves. Therefore, by taking the log, we compress the scale of the variable, reducing skewness and making the data more normally distributed, which improves model performance. This helps address OLS 1 and is in line with the methodology of

Ho (2016). Additionally, it can stabilize variance which helps address for heteroskedasticity where the variance of residuals changes with the magnitude of the variable.

3.2. Model improvements

In response to the limitations of the simple Linear Regression, specification 2 takes the log of CDS prices and reserves, and runs the same linear regression with robust standard errors. This regression predicts the percentage effect of a 1% change in reserves on CDS Prices. This remains statistically significant with a p value near 0, however we now have a 1% increase in reserves leading to a 0.15% decrease in the CDS Price. While this is a more economically significant result, it still fails to account for any controls or time invariant characteristics between the 20 observed countries.

Specification 3 incorporates two controls to account for differences in economic factors that may influence CDS prices. GDP was introduced into the regression to account for the different baseline levels of a country's risk. Large and economically successful countries are less likely to default. This will be reflected in a decrease in the price of that country's CDS. We therefore isolate and remove this variable from the estimation of the desired relationship. As suggested by Akdogan (2019), we also introduced trade surplus in percentage as a second control. Trade surplus will lead to a rise in foreign currency reserves which is not the effect we are measuring. Instead, we want to isolate the effects of Government intervention into reserves. Controlling for trade surplus allows us to remove this variable. Finally, the regression was changed to fixed effects to account for differences in the observed countries that we were unable to control. The equation for the regression with robust standard errors is shown below:

$$\ln(CDS)_{i,t} = \beta_0 + \alpha_i + \beta_1 \ln(Reserves)_{i,t} + \beta_2(GDP)_{i,t} + \beta_3(TradeSurplus)_{i,t} + e_{i,t}$$

The results show that an increase in foreign currency reserves by 1% leads to a decrease in CDS prices by 0.5579%. This was statistically significant with a t-statistic of 3.37. We can therefore reject the null (H_0: $\beta_1 = 0$) that there is no link between foreign currency reserves and CDS prices. This result is now not only statistically significant but also economically significant. These results suggest that a country can reduce CDS by nearly 5.6% by increasing their foreign currency reserves by 10%. Both controls on the other hand had a t-statistics of under 1, meaning those estimations were not statistically significant. Nevertheless, we still expect these controls to

be important to include as otherwise, based on the theoretical background provided by Ho (2016), the model would have omitted variable bias.

In attempting to apply this to actual policy, one important question is whether this effect of reserves is different depending on whether you are in a developing country. Subsequently, in specification 4, we include an interaction term between reserves and a developing nation dummy variable. This regression with robust standard errors estimates that an increase in foreign currency reserves by 1% leads to a decrease in CDS prices by 0.6552%. Since this is the result when the interaction is turned off, this represents the effect for a developed nation. With a t-value of 3.79, it was still statistically significant. The interaction term was estimated at 0.2026, meaning that a 1% increase in reserves for a developing nation would decrease CDS prices by 0.4526. While this shows that the negative relationship holds for both developing and developed countries, it does suggest that the effect is stronger for the latter. The limitation of this result is that the t-value for the interaction term is only 0.65, meaning that we cannot reach conclusive results. We expect that the lack of statistical significance is coming from our small sample size. To improve this, we would expand the scope of this investigation to include more countries.

We hypothesize that developing countries may be less susceptible to changes in CDS since there other non-economic factors leading to their high perceived risk. While foreign currency reserves portray economic strength and stability, it would do nothing if CDS prices are high due to political instability or corruption.

4. Limitations of Results

There were four key limitations we identified throughout the research process. The first limitation was in identifying central bank intervention. As mentioned in the introduction, central banks are purposefully discrete with interventions, thus making it difficult to analyze their impact on markets. Unfortunately, although Uz Akdogan (2019) clearly identifies central bank intervention utilizing foreign currency reserves, the methods utilized to isolate such interventions exceed the complexity of this paper. This means that although we were unable to properly identify central bank interventions, we were able to use reserves as a proxy for interventions.

Secondly, Can et al. (2023) utilizes various econometric methods to find that Turkish CDS spreads unidirectionally influence foreign currency reserves. These findings contradict the

economic theory behind this paper and present a glaring limitation to this paper. Thus, to improve upon our research we should trace the relationship's directionality.

Thirdly, there are clear issues of external validity. From a geographical perspective, there is no guarantee that the found relationship would extend beyond the 20 sampled countries. Additionally, the period from 2015 to 2023 featured a global pandemic, potentially altering results from 2020 onwards. These provide issues to the applicability of our findings.

Lastly, there is clear selection bias embedded in the data analyzed. This is due to a problem of data availability, where we were only able to work with the countries that had CDS, foreign reserves, GDP and trade surplus data in the period examined. Furthermore, the accessible data for GDP, trade surplus, and foreign reserves were not in a daily format. Our solution to duplicate the data, matching the daily frequency of the CDS data, is a possible source of statistical bias.

5. Conclusion

This paper assesses the correlation between foreign currency reserves and sovereign credit default swaps (CDS). This paper uses the foundations set by Ho (2016) and Akçelik & Fendoğlu (2019) to find similar results when incorporating modern data on both emerging and advanced economies. We reach the conclusion that an increase in foreign reserves has a negative effect on premium prices, specifically with a correlation of -0.558 when accounting for controls. This result aligns with economic theory, which suggests that increasing foreign reserves signals stability to investors, thereby reducing the expected premiums on swaps. Additionally, when we introduced an interaction term for developing nations, we found that the relationship between reserves and swap premiums was stronger for developed countries than for developing ones. While this study does not directly contradict Can et al. (2023) and Kartal et al. (2023), as both papers dealt exclusively with Turkey, it does provide a theoretically different perspective to both papers.

With the results of this study, we can recommend to policymakers and central bankers that increasing foreign currency reserves will improve investors perception of their country's risk. Thus, for lawmakers looking to reduce debt servicing costs, it might be worthwhile to increase foreign currency reserves.

6. References

Akçelik, Fatih, and Salih Fendoğlu. “Research Notes in Economics,” 2019.

Bernoth, Kerstin, and Helmut Herwartz. “Exchange Rates, Foreign Currency Exposure and Sovereign Risk.” *Journal of International Money and Finance* 117 (October 1, 2021): 102454. https://doi.org/10.1016/j.jimonfin.2021.102454.

Can, Ali, Fatih Güzel, and Melek Acar. *The Relationship Between Central Bank Reserves and Credit Default Swap Premiums: An Empirical Analysis on Turkey*, 2023.

Coffey, Peter, and Robert J. Riley. *Reform of the International Institutions: The IMF, World Bank and the WTO*. Edward Elgar Publishing, 2006. https://doi.org/10.4337/9781847202888.

Coudert, Virginie, and Valérie Mignon. “The ‘Forward Premium Puzzle’ and the Sovereign Default Risk.” *Journal of International Money and Finance* 32 (February 1, 2013): 491–511. https://doi.org/10.1016/j.jimonfin.2012.05.025.

Foroni, Claudia, Francesco Ravazzolo, and Barbara Sadaba. “Assessing the Predictive Ability of Sovereign Default Risk on Exchange Rate Returns.” *Journal of International Money and Finance* 81 (March 1, 2018): 242–64. https://doi.org/10.1016/j.jimonfin.2017.12.001.

FRED. 2023. "FRED Economic Data." https://fred.stlouisfed.org/

Friedman, Milton. *Essays in Positive Economics*. University of Chicago Press, 1953.

Ho, Sy Hoa. “Long and Short-Runs Determinants of the Sovereign CDS Spread in Emerging Countries.” *Research in International Business and Finance* 36 (January 2016): 579–90. https://doi.org/10.1016/j.ribaf.2015.07.001.

Investing. 2023. "Credit Default Swap Rates (CDS)." https://ca.investing.com/rates-bonds/world-cds?

Kartal, Mustafa Tevfik, Talat Ulussever, Ugur Korkut Pata, and Serpil Kılıç Depren. “Dynamic Link between Central Bank Reserves, Credit Default Swap Spreads, and Foreign Exchange Rates: Evidence from Turkey by Time Series Econometrics.” *Heliyon* 9, no. 5 (May 1, 2023): e16392. https://doi.org/10.1016/j.heliyon.2023.e16392.

Longstaff, Francis A, Jun Pan, Lasse H Pedersen, and Kenneth J Singleton. “How Sovereign Is Sovereign Credit Risk?” *American Economic Journal: Macroeconomics* 3, no. 2 (April 1, 2011): 75–103. https://doi.org/10.1257/mac.3.2.75.

Pelizzon, Loriana, Marti G. Subrahmanyam, Davide Tomio, and Jun Uno. “Sovereign Credit Risk, Liquidity, and European Central Bank Intervention: Deus Ex Machina?” *Journal of Financial Economics* 122, no. 1 (October 1, 2016): 86–115. https://doi.org/10.1016/j.jfineco.2016.06.001.

Rodríguez, Iván M., Krishnan Dandapani, and Edward R. Lawrence. “Measuring Sovereign Risk: Are CDS Spreads Better than Sovereign Credit Ratings?” *Financial Management* 48, no. 1 (2019): 229–56.

Trading Economics. 2023. "Foreign Exchange Reserves." https://tradingeconomics.com/country-list/foreign-exchange-reserves

Uz Akdogan, Idil. “Understanding the Dynamics of Foreign Reserve Management: The Central Bank Intervention Policy and the Exchange Rate Fundamentals.” *International Economics* 161 (May 1, 2020): 41–55. https://doi.org/10.1016/j.inteco.2019.11.002.

Wang, Alan T., and Chin-Chia Liang. “Exchange Rates, Credit Default Swaps and Market Volatility of Emerging Markets: Panel CS-ARDL Approach.” *Borsa Istanbul Review* 24, no. 1 (January 1, 2024): 176–86. https://doi.org/10.1016/j.bir.2023.12.001.

World Bank. 2024. *Export and Import Data*. Accessed December 5, 2024. https://databank.worldbank.org/export-and-import-data/id/42cf7c81.

Tables & Figures

Table 1: Summary Statistics

	Mean	Std. dev.	Min	Max
Price	118.58	152.81	5.46	1836.25
Date	21712.05	853.48	20240	23191
Reserves	501829.2	781928.70	15537	3711143
GDP	3.31E+12	5.02E+12	2.48E+11	2.77E+13
Trade Surplus	0.444	3.860	-12.8	16.93
Developing	0.455	0.498	0	1
Country Group	10.580	5.746	1	20
Log Reserves	12.279	1.241	9.651	15.127
Log Prices	4.200	1.076	1.697	7.515
Interaction	5.630	6.248	0	15.127
Observations	40,129			

Table 2: Regression Analysis of CDS Price and Forex Reserves

	(1) Linear Regression	(2) Logarithmic Linear Regression	(3) Logarithmic Fixed Effects Regression	(4) Interaction + Regression
Reserves (bn)	-0.03			
	(6.08E-04)			
Log Reserves		-0.1543	-0.5579	-0.6553
		(0.0043)	(0.1657)	(0.1730)
GDP			-3.88E-14	-3.86E-14
			(4.01E-14)	(3.95E-14)
Trade Surplus			-0.0063	-0.0069
			(0.0192)	(0.0183)
Interaction				0.2027
				(0.3125)
Constant	135.733	6.095	11.182	11.236
	(1.003)	(0.056)	(2.010)	(1.878)
Fixed Effects	No	No	Yes	Yes
R^2	0.0306	0.0317	0.0761	0.5897
Observations	40,129	40,129	40,129	40,129
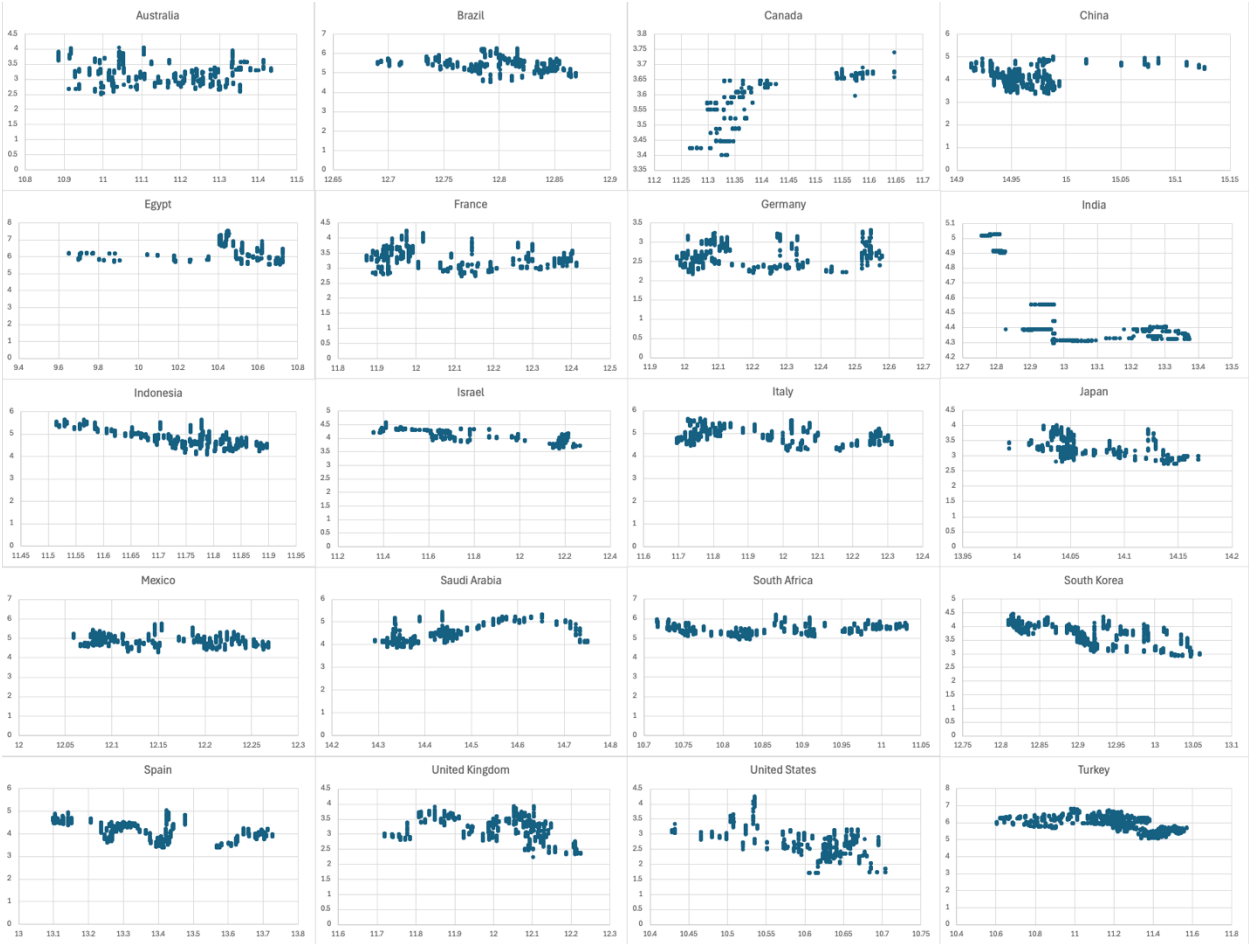
Figure 1- Country Data: ln(Forex Reserves) by ln(CDS Price)